TSA 2019

Problem Solving

Questions taken from the TSA Oxford 2019 Section 1 question paper

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2019 paper, e.g. **1 (Q3)**.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

Score: _____ / 25

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1 (Q2) Each region in the country of Bolandia sets its own tax rate which applies to the whole of an individual's annual income.

Below is a table showing tax information relating to six different people.

name	*region*	*annual income*	*tax rate*
Tim	North	$10 000	30%
Neil	Central	$20 000	25%
Veronica	West	$150 000	3%
David	North West	$50 000	6%
Amy	East	$80 000	5%
Roz	Frontier	$12 000	30%

Who pays the same amount of tax in a year as Tim?

A Neil

B Veronica

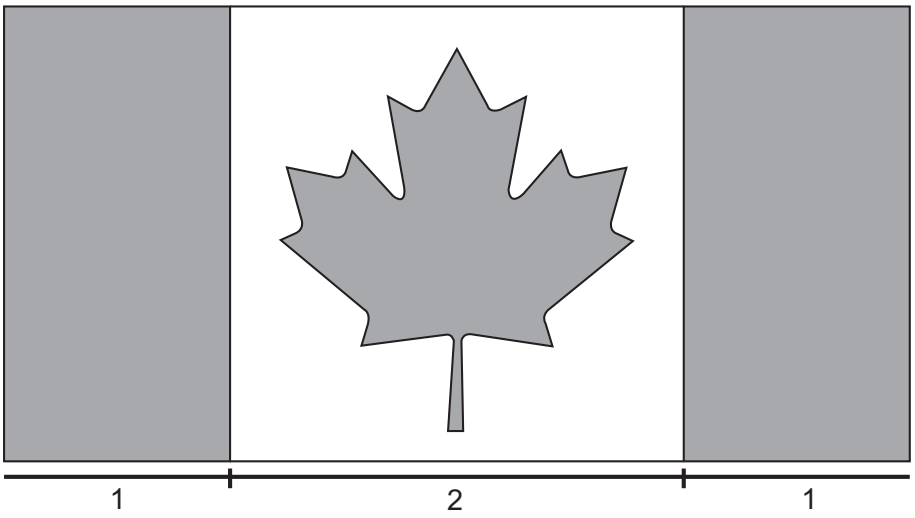
C David

D Amy

E Roz

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2 (Q6) Below is a greyscale image of the flag of Canada. The flag consists of two red strips on the left and the right, a white square in the middle, and a red maple leaf in the centre.



[diagram not to scale]

The width of the white square is twice as large as the width of a red strip. The red maple leaf occupies a quarter of the square in the middle.

What is the ratio of red to white in the flag?

A 1 : 1

B 2 : 1

C 3 : 2

D 5 : 3

E 5 : 4

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3 (Q7) The table below shows the breakdown of marks awarded to students taking an exam.

student	*first examiner*	*first mark*	*second examiner*	*second mark*	*final mark*
Arnold Jones	Ms Rosenberg	80	Mr Robson	70	75
Betty Markham	Mr Kettering	90	Mrs De Vere	60	75
Clara Bird	Ms Nolan	60	Mr Robson	50	55
Cora Smith	Mr Kettering	80	Ms Rosenberg	80	80
Dave Howard	Mrs De Vere	90	Ms Rosenberg	90	90
Dennis Garton	Mr Kettering	80	Ms Nolan	70	75
Hillary Gordon	Mr Robson	50	Mrs De Vere	90	70
Janine Turner	Ms Rosenberg	80	Ms Nolan	90	85
Laurence Rothman	Ms Nolan	60	Mrs De Vere	50	55

The mark scheme is as follows: 90 = excellent; 80 = good; 70 = satisfactory; 60 = acceptable; 50 = fail. The final mark is the average between the marks given by the first and second examiners.

Which pair of examiners disagreed the most on the marks they awarded a student?

A Mr Kettering and Mrs De Vere

B Mr Robson and Ms Nolan

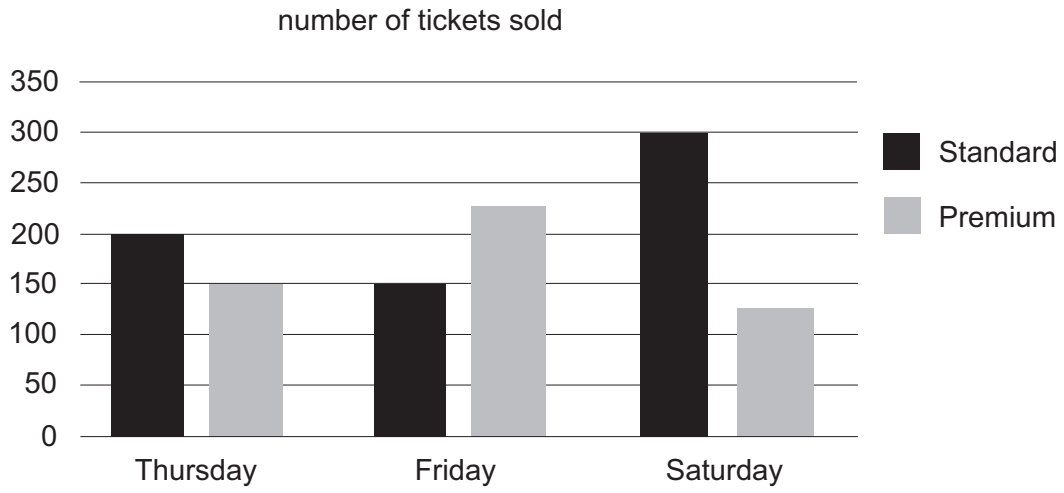
C Ms Rosenberg and Ms Nolan

D Mr Kettering and Ms Rosenberg

E Mrs De Vere and Mr Robson

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4 (Q8) The bar chart shows the total numbers of tickets sold for a concert over three nights.



Premium tickets are more expensive than Standard tickets.

Which one of the following statements is correct?

A The highest total sales income was for Thursday.

B The highest total sales income was for Friday.

C The highest total sales income could have been for Thursday or Friday.

D The highest total sales income could have been for Thursday or Saturday.

E The highest total sales income could have been for Friday or Saturday.

5 (Q12) When riding my bike in the dark, I use two rear bike lamps. Both of the lamps are in 'flashing mode', which means that they repeatedly emit light for a set period of time at set intervals. The reasons for using two lights are that it reduces the amount of time when neither of the lamps emit light, and that if one of the lamps stops working, other road users can still see me on the road. One of the lamps shines for one second and goes dark for one second repeatedly; the other lamp shines for two seconds and goes dark for two seconds repeatedly. I always turn both lamps on simultaneously at the start of my ride.

For how many seconds in each minute do **neither** of the lamps shine when they are in use?

A 10

B 12

C 15

D 20

E 30

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6 (Q13) The following table compares five selected commuting routes in different European countries with respect to the length of the journey and the monthly cost of a season ticket as a percentage of average monthly wages.

comparison of selected commuting routes			
journey	*country*	*length* (km)	*cost as a percentage of average monthly wages*
Luton–London	UK	56	14
Brandenburg–Berlin	Germany	69	8
Karis–Helsinki	Finland	75	6
Uppsala–Stockholm	Sweden	72	6
Drogheda–Dublin	Ireland	46	5

On average, the commuters of which route spend the lowest proportion of their wages per kilometre of the route?

A Luton–London

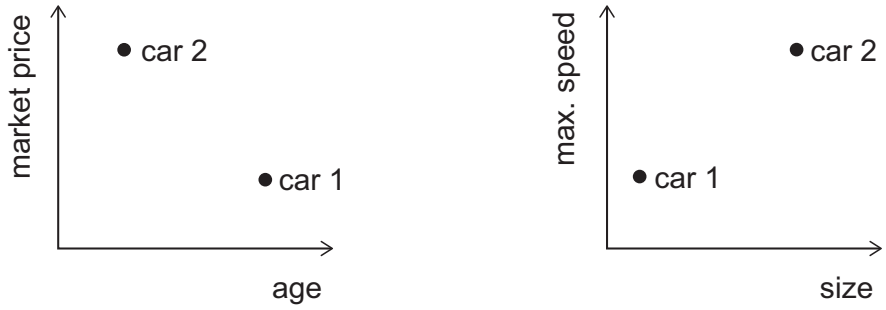
B Brandenburg–Berlin

C Karis–Helsinki

D Uppsala–Stockholm

E Drogheda–Dublin

7 (Q14) Peter has drawn two diagrams showing information about the two cars owned by his parents:



Which one of the following statements is **not** correct?

A The more expensive car is newer.

B The cheaper car is bigger.

C The older car is slower.

D The newer car is bigger.

E The slower car is smaller.

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8 (Q18) Three friends are training to run a marathon. Part of this training involves running steadily round a track in the local park. They are all at different stages in their training schedules so they run at different speeds. Alec completes each lap in 3 minutes, Barry takes 5 minutes and Colin takes 6 minutes.

They all leave the start line at the same time.

If they each maintain their running speed throughout the session, how many more laps will Alec have completed compared to Barry when the three of them next cross the start line together?

A 2

B 4

C 5

D 6

E 10

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9 (Q19) The table below shows the amount of waste recycled each year, in thousands of tonnes, over a three-year period in 20 countries.

country	*recycling year 1*	*recycling year 2*	*recycling year 3*
Ambrosia	52	64	65
Buranda	60	72	80
Carbonda	20	27	40
Elbonia	32	34	40
Friponia	36	39	45
Grandia	20	35	60
Hoollasia	50	56	60
Ishlapia	33	41	44
Jackaran	36	44	48
Killumpa	40	55	70
Lateralia	50	57	60
Moodle	80	98	90
Nalpha	32	44	48
Oritan	72	81	90
Phillim	66	69	70
Quaratus	80	90	100
Rutalia	75	76	90
Samalam	88	96	99
Tuddle	50	56	60
Ukliam	50	56	60

What is the biggest percentage increase in recycling in the table from year 1 to year 3?

A 50%

B 75%

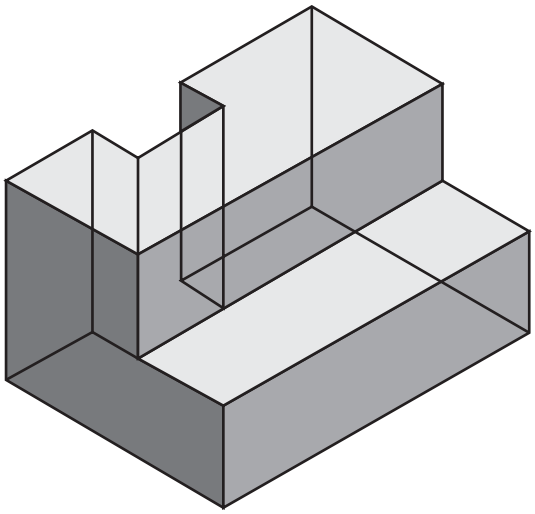
C 100%

D 150%

E 200%

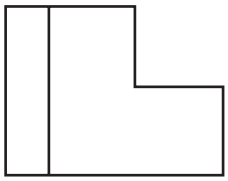
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10 (Q20) The see-through glass block below can be viewed from six sides, including from above and below.

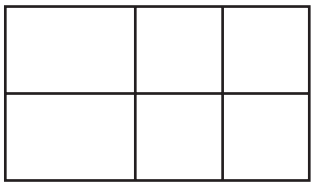


Which one of the following is **not** a possible view?

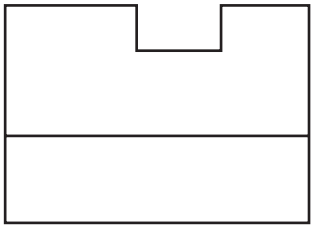
A



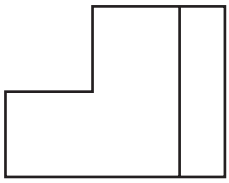
B



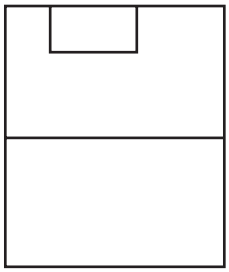
C



D

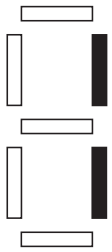


E



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11 (Q24) A digital display consists of seven segments which light up in different combinations to produce digits. For example, to display the digit '1' the two right-hand segments light up as shown:



The display is used to show, in sequence, each of the digits 1 to 9.

What is the maximum and minimum number of times that any one segment lights up during this sequence?

	maximum	*minimum*
A	7	4
B	7	5
C	8	3
D	8	5
E	9	4

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12 (Q25) Sandi is planning a trip to the old city of Pordson. She wants to stay for four nights in order to visit all of the attractions, and one of the nights must be the Saturday so that she can attend a special music concert.

The table below shows the prices per night and details of the deals for the five main hotels in Pordson.

Hotel	*Mon*	*Tues*	*Weds*	*Thurs*	*Fri*	*Sat*	*Sun*		*Deal*
Grand	€60	€70	€70	€70	€50	€60	€60		3F2
Majestic	€40	€90	€70	€70	€40	€90	€40		SFSHP
Palace	€60	€90	€90	€70	€50	€90	€80		50%
Palms	€60	€80	€80	€50	€40	€60	€50		4FP3
Plaza	€70	€90	€90	€60	€50	€70	€50		CBK

Deals: 3F2 Stay 2 nights and get the 3rd night free
SFSHP Stay Friday and get the Saturday night half price.
50% 50% discount on all stays of 3 or more nights.
4FP3 Stay 4 nights and get the cheapest night free.
CBK €50 discount on stays of 3 or more nights.

What is the minimum that Sandi will need to spend on a hotel bill for her stay in Pordson?

A €110

B €120

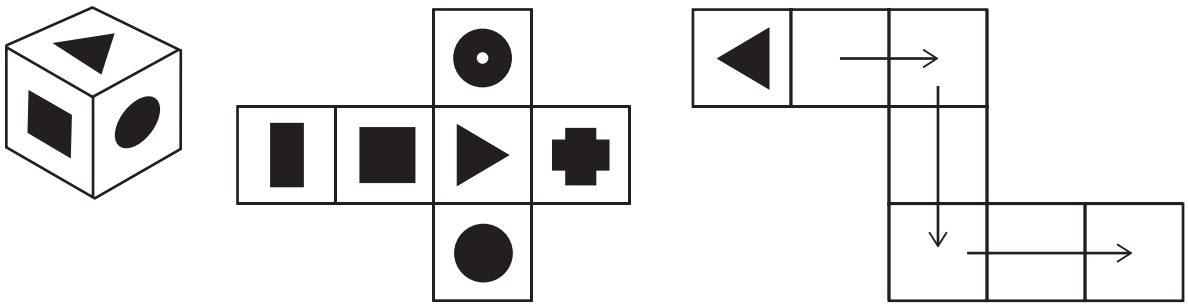
C €140

D €145

E €160

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13 (Q26) A solid cube has 6 different shapes drawn on each of its 6 faces. A plan view showing all the shapes is given below. It is placed on the path drawn below with the triangular shape facing up. The cube is rolled, one face at a time, forward along the route shown, from start position until it reaches the end position.



Which one of the following shapes is facing up when it reaches the end position?

A



B



C



D



E



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14 (Q30) Alex and Sue are playing a game with a set of nine cards, numbered from 1 to 9. A 3 × 3 grid of cards is dealt onto the table. The cards for the next round have just been dealt and are as shown below.

5	1	8
3	6	2
9	7	4

The first player chooses two cards and scores the difference between the two numbers. The second player then chooses two cards from the seven that are left and scores the average of the two numbers. The highest score wins the round. Both players must choose cards that are either horizontally or vertically next to each other in the grid.

For this round of the game Alex has to decide whether to play first or second. She wants to make sure that she wins the round. If she has more than one possible winning move then she wants to achieve the highest score possible.

What should Alex choose to do?

A play first and choose the 1 and 8

B play first and choose the 3 and 9

C play first and choose the 6 and 7

D play first and choose the 7 and 9

E play second

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15 (Q31) The single ferry to Pepper Island starts each day from mainland Seatown and crosses to the island port of Kaysville. It runs back and forth from 9:30 am to 4:10 pm on Saturdays, Sundays and Bank Holidays, and from 10:30 am to 2:20 pm from Monday to Friday.

MAY

M	T	W	T	F	S	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

highlighted days = Bank Holidays

Each crossing takes 50 minutes. Between crossings, there is a 10-minute stopover for cleaning and refuelling. This is extended to 20 minutes on Saturdays, Sundays and Bank Holidays to allow for the increased cleaning requirements.

What is the maximum number of crossings the ferry can make in May?

A 140

B 144

C 154

D 166

E 186

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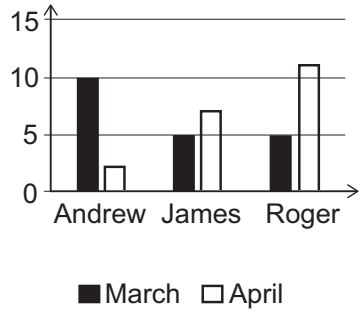
16 (Q32) In March, 20 people voted for the position of chair of a committee. There were three potential candidates: Andrew, James and Roger.

In April, 11 of these people changed their votes, as follows:

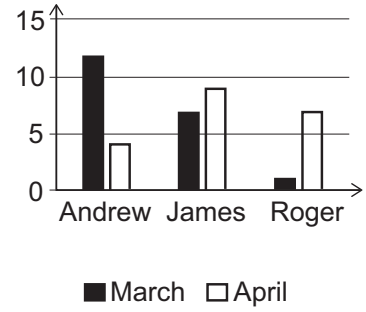
3 people changed their vote from Andrew to James
2 people changed their vote from James to Roger
5 people changed their vote from Andrew to Roger
1 person changed their vote from Roger to James

Which one of these bar charts could **not** show the number of votes for the three candidates in March and April?

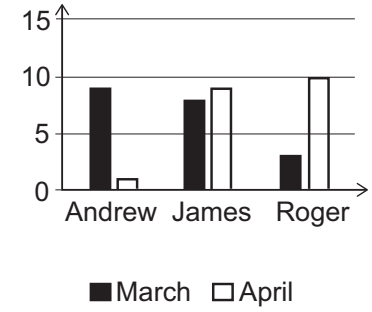
A



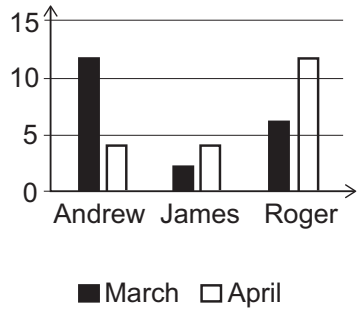
B



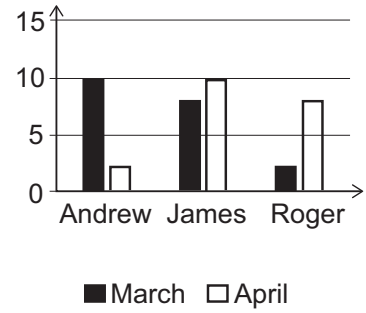
C



D

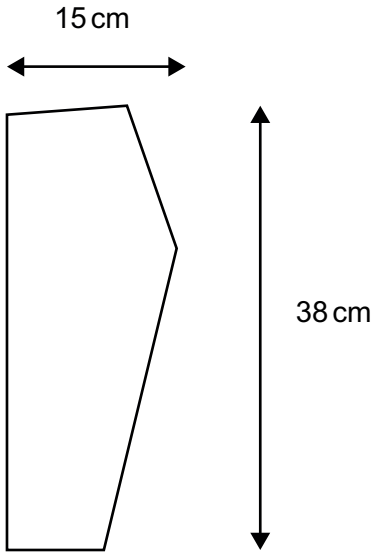


E



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17 (Q36) Nathan is making a pair of trousers for his young son. The diagram shows the pattern for one piece. Four such pieces are needed.



[diagram not to scale]

For simplicity he assumes that he will need a 15 cm × 38 cm rectangle for each piece. The material comes in three widths with prices as shown:

width	*price / metre*
100 cm	£2.00
120 cm	£2.20
160 cm	£2.80

Material may be bought in lengths which are exact multiples of 10 cm.

What is the lowest price he can pay?

A 42p

B 56p

C 60p

D 66p

E 80p

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18 (Q37) Arthur plans to modernise his bathroom. His design requires a 1600 mm × 700 mm bath. He wants side grips to help him get out of the bath, and because he doesn't want to choose the taps before ordering the bath he must have one which doesn't have pre-drilled tap holes.

A list from the bath catalogue is shown below:

name	*material*	*length*	*width*	*height*	*side-grips*	*pre-drilled tap holes*	*price* (£)
Balmoral	acrylic	1600	750	590	yes	no	159
Carola	acrylic	1600	700	610	yes	no	130
Dalton	steel	1600	700	580	no	no	112
Europa	steel	1600	700	590	yes	no	126
Gamma	steel	1600	700	580	yes	yes	135
Oporto	acrylic	1600	700	585	no	no	115
Polar	steel	1500	700	600	yes	no	120
Rocca	acrylic	1600	700	590	no	no	126
Sierra	steel	1600	700	580	yes	no	128
Victory	acrylic	1600	700	580	yes	no	149

How much will Arthur save by choosing the cheapest steel bath that fits his requirements rather than the cheapest acrylic bath that fits his requirements?

A £2

B £3

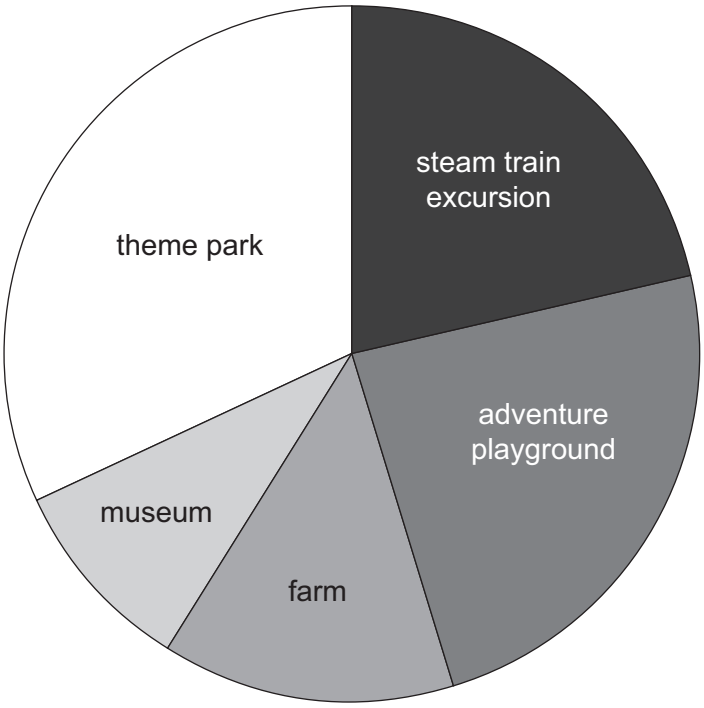
C £4

D £10

E £18

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19 (Q38) The Head of Year 6 is planning an end-of-term treat for her students. The pie chart represents the numbers of votes cast by the children for each of the five possible outings.

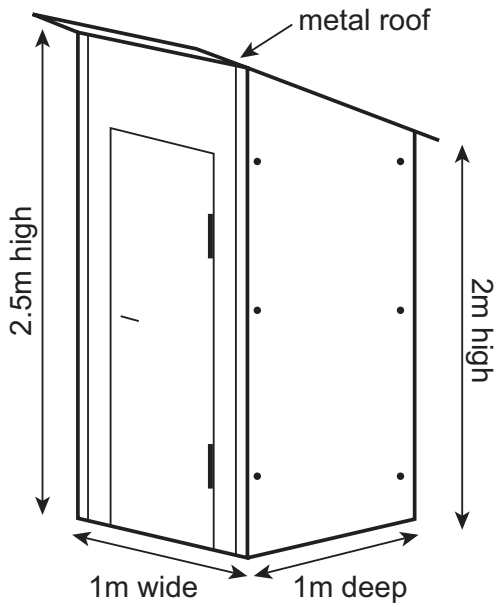


Which one of the following rows (**A–E**) could show the actual number of votes cast by the children for each of the options?

	steam train excursion	*adventure playground*	*farm*	*museum*	*theme park*	*total*
A	35	45	30	15	55	180
B	24	28	20	16	32	120
C	45	55	20	15	65	200
D	22	28	15	15	40	120
E	32	36	20	14	48	150

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20 (Q42) The drawing below shows an allotment shed.



[diagram not to scale]

The four outside wood surfaces of the shed, including the door, need to be painted with three coats of paint. It takes 4 minutes to paint 1 square metre.

I have two identical sheds to paint and will take just one 20-minute tea break.

How long will it take to paint the two sheds?

A 36 minutes

B 1 hour and 48 minutes

C 2 hours and 8 minutes

D 3 hours and 36 minutes

E 3 hours and 56 minutes

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21 (Q43) A school party consisting of 4 teachers and 20 students is visiting the theatre. Of the students, 8 are aged 15 years and 12 are aged 16 years. The following table shows the entry prices to the theatre.

category	*price*
Adult (16 years and over)	$30
Child (5–15 years)	$20
Under 5 years	free
Family ticket (2 adults and up to 2 children under 16 years)	$90
Group ticket (8 people including at least 2 children under 16 years)	$175
Supergroup ticket (any 8 people)	$200

What is the least amount that the school party must pay for entry to the theatre?

A $525

B $530

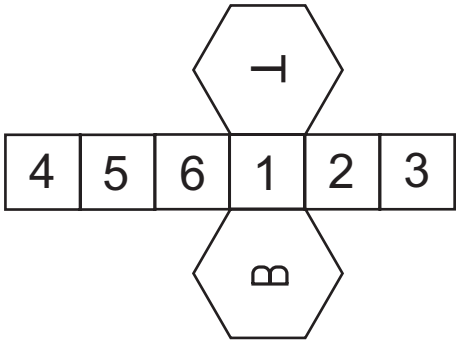
C $545

D $550

E $560

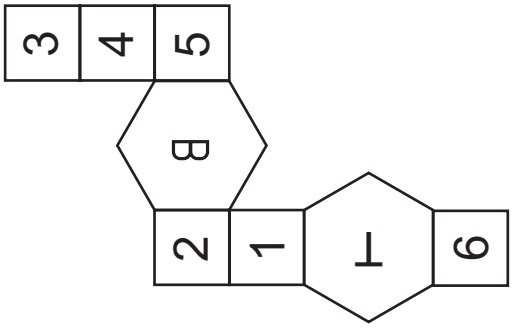
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22 (Q44) Consider the following net, which forms a hexagonal prism:

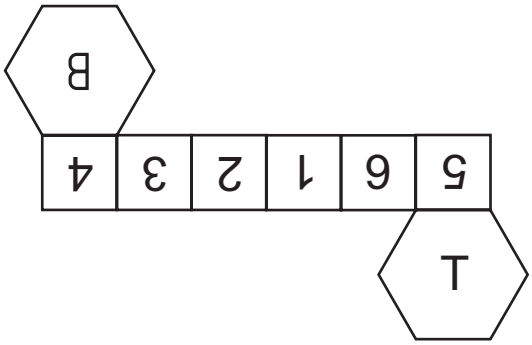


Which one of the following nets will form an identical prism?

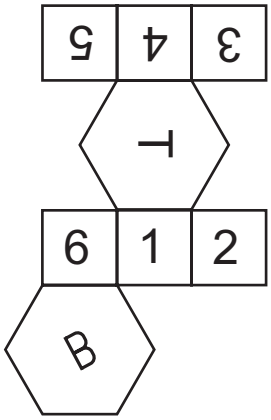
A



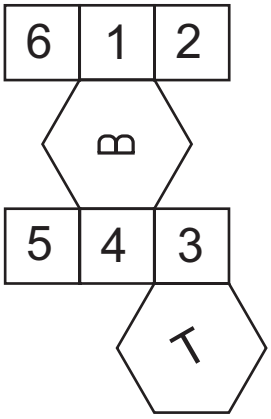
B



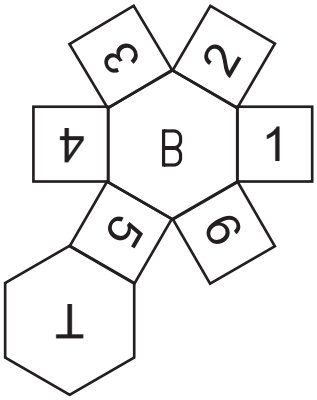
C



D



E



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23 (Q48) Sourdough, an ingredient of many bread products, is a fermented mixture of flour and water. At 8:00 am on Monday morning, I start with a mixture of 50 g flour and 50 g water. Each morning at 8:00 am I double the weight of the sourdough by adding equal weights of flour and water.
I know that 1 g of the water content evaporates every hour.

The bread recipe that I would like to use requires 500 g of sourdough. I also want at least 50 g of sourdough left over so that I can continue growing it for next week's bread.

On which day am I going to have the necessary amount of sourdough?

A Tuesday

B Wednesday

C Thursday

D Friday

E Saturday

24 (Q49) A 250 g bag of mixed nuts lists the contents on the packet as follows:

nut	*content*
almonds	20-30%
brazil nuts	10-20%
hazelnuts	5-15%
peanuts	30-40%
pecans	10-20%

As they are my favourite, I eat all the almonds in the packet and then re-weigh it. I find that the almond content was the minimum it could have been whilst being consistent with the labelling.

What is the largest possible peanut content of the nuts remaining in the bag after I have eaten the almonds?

A 40%

B 48%

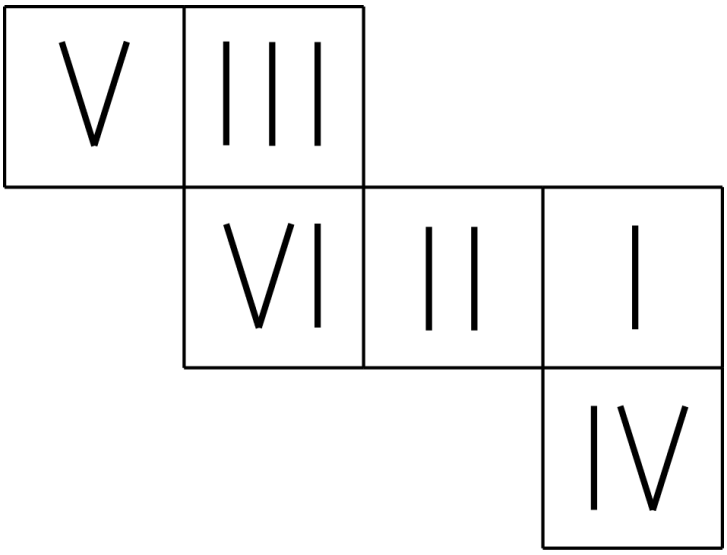
C 50%

D 57%

E 60%

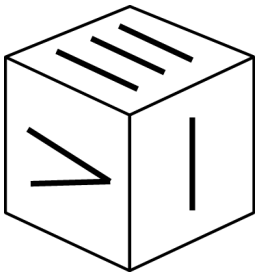
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25 (Q50) The net below can be folded to make a Roman numerical dice.

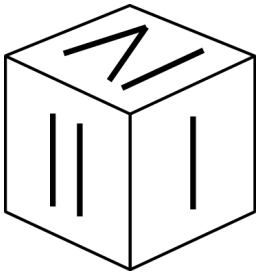


Which one of the following dice could have been made as a result of folding the net?

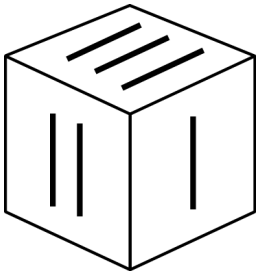
A



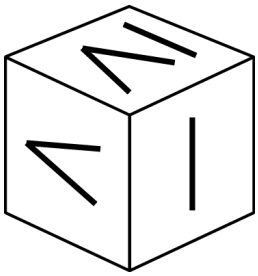
B



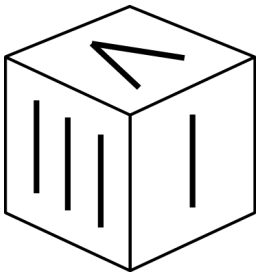
C



D



E



END OF TEST

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